MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS

TYPE 8435 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Diode, hydrogen

See figure 1

Mounting position: Horizontal or vertical (base down)

Weight: 30 ounces nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Ib	tk	Fault electrode current (0.1 sec max)	TA
Unit:	Vac	kv	a	Adc	sec	a	°C
Rectifier:							
Maximum:	5.25	20 (Notes 1 & 2)	8	1.8 (Note 1)	---	60	+75
Minimum:	4.75	---	---	---	180	---	-55
TEST CONDITIONS:	5.0	22	8	2	180	---	---

GENERAL:

Qualification - Required

1/ See note 8

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Heater current		0.65	II	If	9.5	13.8	Aac
3249	Rectifier operation (1)	Ef = 4.75 Vac; t = 5 minutes (see notes 3, 4, and 5)	0.65	II	---	---	---	---
3249	Rectifier operation (2)	Ef = 5.25 Vac; t = 5 minutes (see notes 3, 4, and 5)	0.65	II	---	---	---	---
3265	Hydrogen diode ionization voltage	See notes 6 and 7	0.65	II	Ez	---	100	v
3204	Voltage drop	See notes 6 and 7	0.65	II	etd	---	60	v
	Quality conformance inspection, part 3							
----	Life-test provisions	Group C; t = 1,000 hours (see notes 4 and 5)	---	---	---	---	---	---
----	Life-test end points							
3249	Rectifier operation (1)		---	---	---	---	---	---
3204	Voltage drop		---	---	etd	---	80	v
	Periodic-check test							
3249	Rectifier operation (3)	t = 60 minutes, TA = 75° C (see notes 3 and 4)	---	---	---	---	---	---

NOTES:

1. Maximum current and voltage ratings do not apply simultaneously. See figure 2 for current versus voltage ratings.
2. Whenever possible it is desirable to apply anode voltage gradually. Apply 50 percent of voltage at end of tk period and raise to maximum in 15 seconds.
3. Three arc backs are permitted prior to an uninterrupted operation run of specified duration during which no sparking, flicker, or abnormal glow discharge shall occur.
4. This test shall be performed with an airflow of 50 cfm directed down on top of the radiator.
5. At t = 180, apply 11 kv inverse voltage and raise to 22 kv in approximately 15 seconds.
6. Use low voltage (110 to 660 volts) 60 Hz, half-wave rectifier circuit in which Io = 1.8 Adc and Ib = 8 a.
7. The value of hydrogen diode ionization voltage and voltage drop shall be determined by means of a calibrated oscilloscope connected between anode and cathode after the tube has conducted for 120 seconds (see figure 3).
8. Tube type 8434 is hereby deleted from this specification sheet. Tubes for replacement purposes will be procured to EIA data, Release Number 4396.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 17, 80
DSA - ES

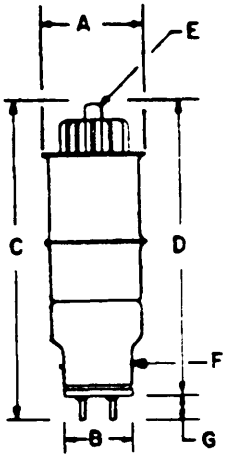
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

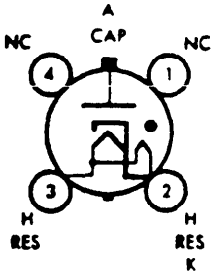
Agent: DSA - ES

(Project 5960-2425-86)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
(B) Quality conformance inspection, part 2		
A		2.975 (75.57)
B	2.187 (55.55)	2.219 (56.36)
C		10.41 (264.41)
D	8.78 (223.01)	9.66 (245.36)
E	Cap: C1 - 5 (EIA)	
F	Base: A4 - 11 (EIA)	
G		.74 (18.80)



(SEE NOTE)
BOTTOM VIEW

NOTE:
The cathode connection shall be made to pin 2 in order not to damage tube.

FIGURE 1. Outline drawing of electron tube type 8435.

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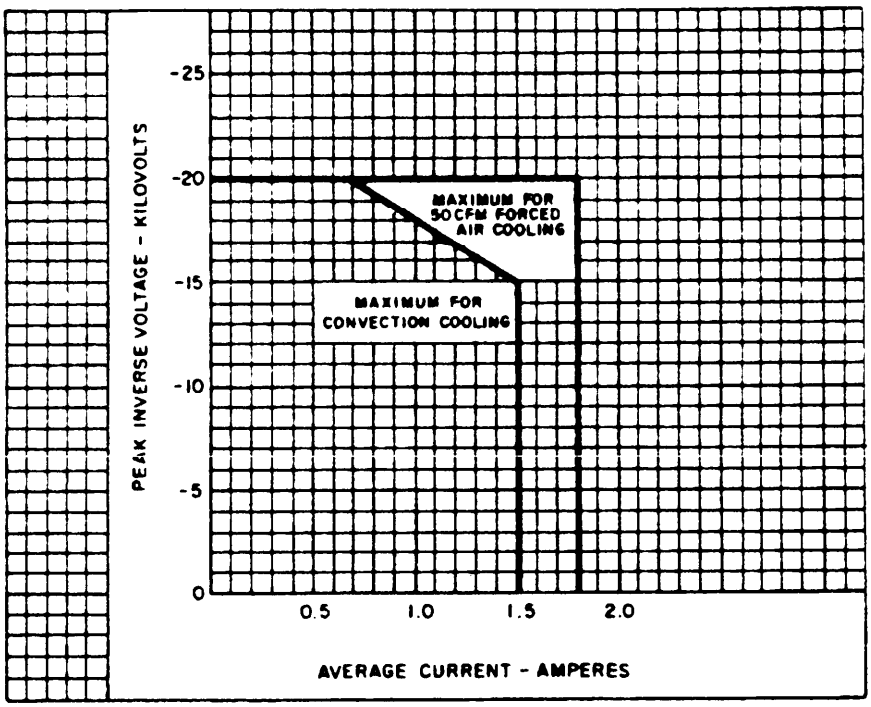


FIGURE 2. Current versus voltage ratings.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R0

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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